A-A-53577
29 July 1988

COMMERCIAL ITEM DESCRIPTION

ELECTROSURGICAL APPARATUS

(Portable, Unblended Current)

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers one type of portable, electrosurgical apparatus, suitable for desiccation, fulguration, and bi-active coagulation of human tissue in office treatment procedures.

Salient characteristics:

Material. Material shall be of the highest quality throughout, suitable in all respects for the purpose intended. The case and housing shall be made of a high impact, high dielectric plastic polymer consisting of acrylonitrile, butadiene, and styrene or a polyphenylene oxide polymer (such as "ABS" or "Noryl" plastic).

Design and construction. The electrosurgical apparatus shall be a portable dual-spark-gap type, adapted for wall mounting or desk and table top use. It shall operate from a 110 to 125 volt, 50 to 60 Hertz alternating current power source. Rated power consumption shall not exceed 150 watts.

High-frequency generator assembly. The generator assembly shall contain the electrical components and circuitry required to produce damped output currents ranging from 500,000 Hz to 2,000,000 Hz. The patient circuit shall be filtered to block passage of low frequency currents. The unit shall be fitted with at least three outlets; one each for "low" and "high" power electro-desiccation or fulguration and two outlets used for bi-active coagulation. These outlets shall be suitably identified and provide jacks for connection of the proper monoterminal or biterminal handled plugs on the front of the generator for selection of the desired surgical technic.

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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The high-frequency generator shall include a power transformer, dual spark gaps, and a single linear calibrated dial control. The dial control shall provide stepless linear output current regulation for each technic, starting at zero output current for zero setting and increasing to the maximum value at the largest dial setting. Sensitivity of the dial control shall be fine enough to provide the necessary current regulation on the low setting for all applicable surgical procedures. Current output shall not drift on any dial location. Calibration shall be performed at the factory and not be required again for the useful life of the instrument. The output current shall be linear, directly proportional to the dial setting, and repeatable for every possible dial setting. The generator assembly shall be enclosed in a high impact case with a removable back plate. Provisions on the back plate shall permit wall mounting the generator assembly or the wall bracket supplied with the unit. The case shall include a handle with retaining arrangements to hold the handle of the electrode set plugged into the unit, when the apparatus is not in use. The power cord shall be a three blade male connector fitting receptacle specified under American National Standards Institute Standard C 73.11. The high-frequency generator output current shall be at least 60 milliamperes (Ma) with a 300 ohm load for the low output, 160 Ma with a 300 ohm load for high output, and 180 Ma with a 300 ohm load for the bi-active output.

Foot switch. The foot switch shall be a normally open floor-type switch, which shall close upon application of foot pressure and energize the high-frequency generator. Foot switch shall be permanently connected to the unit by a two-conductor cord (U.L. listing). The cord shall be secured to case by special bracket and strain relieved internally.

Electrode set, Electro-desiccation-fulguration. The unit shall consist of an insulated single-conductor cord with a monoterminal handle connected at one end, a nominal 6 foot length single conductor cord, and a jack at the other end for plugging into indicated "Low" - "High" - power outlets in the generator. The set shall include 3 electrodes which shall fit into the handle, one small ball tip electrode, one straight needle type electrode, and one straight point electrode.

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Bipolar coagulation. The unit shall have bi-polar coagulation capabilities and all internal case provisions, including indicated outlets for "Bi-polar" coagulation on the generator. Optional accessories such as an insulated two-conductor cord with a bipolar handle connected at one end and two jacks at the other end, and a set of bipolar electrodes for bi-polar coagulation, and other electrodes are not part of this specification, but must be available for the apparatus.

Dielectric strength. The apparatus shall operate properly both prior to and after the dielectric test specified herein. There shall be no evidence of dielectric breakdown during this test.

Identification Marking. Each electrosurgical apparatus shall be permanently and legibly marked with the following information:

Manufacturer's name or registered trademark
Contract or purchase order number
National stock number
Voltage and frequency.

Marking shall be by impressing, stamping, or engraving, or by nameplates permanently attached. Marking secured by adhesive shall not be used.

Fire and casualty hazard.

The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of the Underwriter's Laboratories Inc. Proof shall consist of one of the following:

A. UL label or UL listing of the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the Contracting Officer, upon the latter's request.

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B. Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facility's report and to conduct a facility's survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushing, photographs, wiring diagrams, sketches, and parts list, all appropriately identified. Compliance with this paragraph does not excuse compliance with other requirements of the contract and specifications.

Operational and Service Data Manuals shall be furnished as follows:

A. Information Content: The contractor shall furnish manuals, handbooks and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures of illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures will be written in the English language and all schematics and wiring diagrams will use American electrical and electronics symbols. The manuals will include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name and quantity required. All service notes, service memos, etc., pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as a minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

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Quantities, distribution and scheduled:

Two (2) complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of 2 service manuals need to be supplied to each consignee. However, 2 operation manuals still need to be supplied with each unit.

Workmanship. The electrosurgical apparatus shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One electrosurgical apparatus, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures, and testing (determination by technical means of physical and chemical properties, of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

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First article inspection. First article inspection shall be performed on the one complete electrosurgical apparatus when a first article sample is required. This inspection shall include the examination and tests.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one complete electrosurgical apparatus as specified in this description.

Defects.

Metal fitting not adequately secured to cord.

Handle not free or cracks, ruptures, and deformed sections.

Cord insulation not free of tears, cracks or breaks.

Ends of cord not free of exposed wire at points of entrance into handle or fitting.

Handle interior not free of obstructions to chuck component.

Unit not free of dirt, grease and foreign matter.

Cord not flexible as required.

Container not properly sealed to prevent contamination of inner contents.

Handle and cord color does not contrast with light blue and green surgical drapes.

Sampling for tests. Sampling for tests shall be conducted as follows:

a. First article. Where a first article sample is required, all tests shall be performed on that unit.

b. Acceptance testing. Wher a first article sample has not been required, one unit shall be subjected to the performance test specified herein. If a first article sample has successfully passed all tests, performance test shall not be required at acceptance.

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c. Dimensional test. Sampling for dimensional test shall be in accordance with MIL-Std-105 with an AQL of 1.5 (percent defective) and an inspection level of S-1. The unit of product for tests shall be one electrosurgical apparatus.

Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

Functional test. The unit shall be connected to nominal 115 volt, 50- 60 cycle, alternating-current power source, and shall be operated at each technic using a test medium, such as a coin, small piece of meat, other practical material which visually indicates the intensity of the output current applied, or an electrical measuring instrument. For each technic, the dial control shall be analyzed to determine whether the unit functions properly and the dial control varies the output current.

Dielectric Strength Test. A gradually increasing 60 cycle potential of 2500 volts AC shall be applied between the current-carrying parts of the alternating current 50-60 cycle primary source and the user electrodes, and ground for a period of one minute.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

6515-00-935-9763

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MD

Agent:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4227

PGC 46623

Location:
CID/ELECTROSURGICAL/H16-31

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53577	ELECTROSURGICAL APPARATUS (Portable, Unblended Current)

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Including Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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